

Algebra 1
Unit 6
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Jacqueline is buying grapes and apples from the Walmart produce section. Grapes are \$3 per pound and apples are \$4 per pound. Jacqueline's mother gave her \$20 to spend. To determine the combinations of grapes and apples that she can afford to purchase, Jacqueline wrote the inequality $3x + 4y \leq 20$.

1. Define the variables Jacqueline used based on the context.

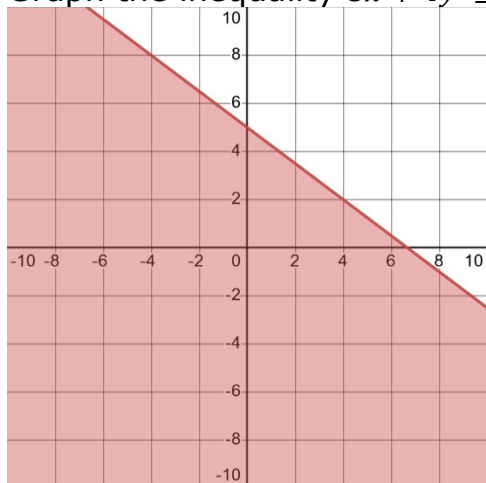
Let x represent the weight of grapes purchased, in pounds.

Let y represent the weight of purchased apples, in pounds.

2. Explain why the inequality sign $\leq$ makes sense in this context.

The inequality sign, $\leq$, in this context makes sense because it means they can spend up to their max, which is \$20.

3. Graph the inequality $3x + 4y \leq 20$.



4. The ordered pair $(3, -4)$ is a solution to the inequality $3x + 4y \leq 20$. Explain why this is not a viable solution in the context of buying grapes and apples.

It is not viable because y represents the weight of apples purchased, which cannot be negative.

5. The ordered pair $(4, 2)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

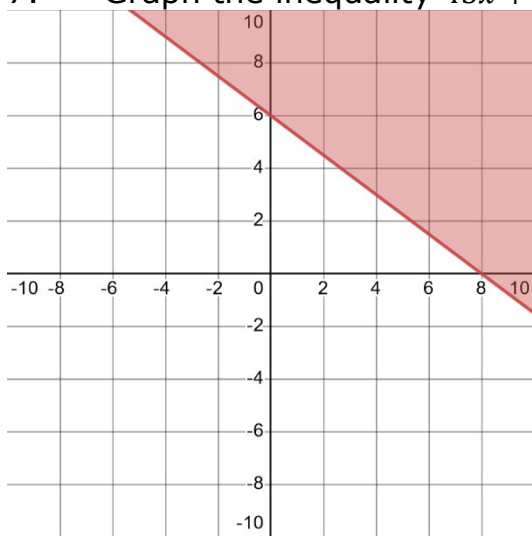
Jacqueline can purchase 4 pounds of grapes and 2 pounds of apples for exactly \$20.

Miguel invested some of his savings in the stock market. He bought 45 shares of a video production company and 60 shares of a clothing company. He hopes to make a combined profit of at least \$360 when he sells his shares. He wrote the inequality $45x + 60y \geq 360$ to represent the situation, where x is the change in value of one share of the video production company and y is the change in value of one share of the clothing company.

6. Explain what it would mean if x or y were a negative number.

The value of the shares will have decreased in value from the time of Miguel's initial investment.

7. Graph the inequality $45x + 60y \geq 360$.



8. Explain what the point $(-2.50, 9.75)$ represents in Miguel's situation.

Shares of the movie stock decreased by \$2.50, but shares of the clothing stock increased by \$9.75, resulting in a combined profit of $45(-2.5) + 60(9.75) = \$472.50$, which is ≥ 360 .

9. Explain why it makes sense in Miguel's context that there are no solutions for which both x and y are both negative.

If both x and y are negative, profit earned will never be positive, let alone equal to \$360.

10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to 2 decimal places.